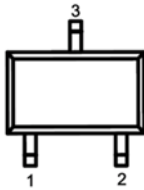
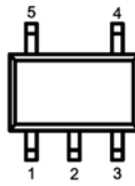


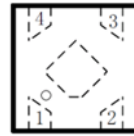
CE6200 Series

■ PIN CONFIGURATION

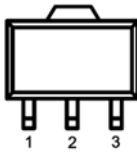
SOT-23-3

SOT-23-5/
SOT-353(SC70-5)

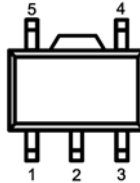
DFN1×1-4



SOT-89-3



SOT-89-5



CE6200AXX (SOT-23-3)

PIN NUMBER				SYMBOL	FUNCTION
M	MA	MC	MY		
1	2	3	3	V _{SS}	Ground
2	1	2	1	V _{OUT}	Output
3	3	1	2	V _{IN}	Power Input Pin

CE6200BXXM/CE6200AXXMF/ CE6200BXXML (SOT-23-5)

PIN NUMBER			SYMBOL	FUNCTION
M	MF	ML		
1	1	5	V _{IN}	Power Input Pin
2	2	2	V _{SS}	Ground
3	—	1	CE	Chip Enable Pin
4	3/4	3	NC	No Connection
5	5	4	V _{OUT}	Output Pin

CE6200BXXU (SOT-353(SC70-5))

PIN NUMBER		SYMBOL	FUNCTION
U			
1		V _{IN}	Power Input Pin
2		V _{SS}	Ground
3		CE	Chip Enable Pin
4		NC	No Connection
5		V _{OUT}	Output Pin

CE6200BXXF (DFN1×1-4)

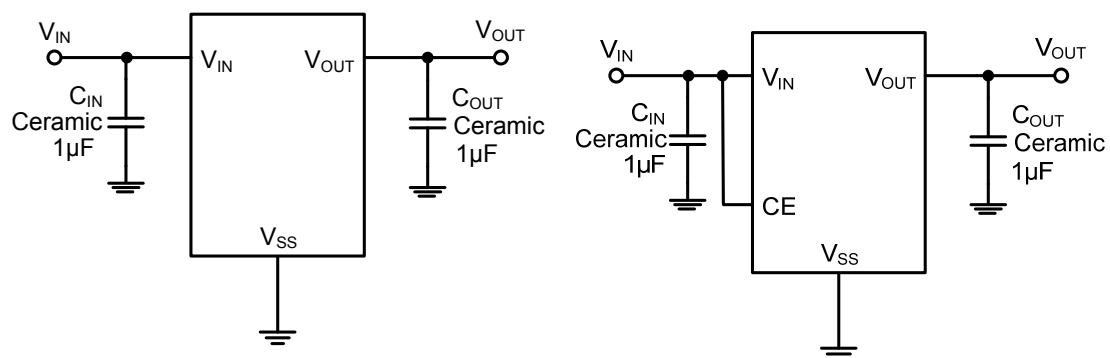
PIN NUMBER	SYMBOL	FUNCTION
F		
1	V _{OUT}	Output Pin
2	V _{SS}	Ground
3	CE	Chip Enable Pin
4	V _{IN}	Power Input Pin
EP	Thermal PAD	Ground

CE6200AXX (SOT-89-3)

PIN NUMBER		SYMBOL	FUNCTION
P	PT		
1	2	V_{SS}	Ground
3	1	V_{OUT}	Output
2	3	V_{IN}	Power Input Pin

CE6200BXXP (SOT-89-5)

PIN NUMBER	SYMBOL	FUNCTION
P		
1	V _{OUT}	Output Pin
2	V _{SS}	Ground
3	NC	No Connection
4	CE	Chip Enable Pin
5	V _{IN}	Power Input Pin

■ TYPICAL APPLICATION

■ ABSOLUTE MAXIMUM RATINGS⁽¹⁾

(Unless otherwise specified, $T_A=25^{\circ}\text{C}$)

PARAMETER		SYMBOL	RATINGS	UNITS
Input Voltage ⁽²⁾		V_{IN}	-0.3~7	V
Output Voltage ⁽²⁾		V_{OUT}	- 0.3~ $V_{IN}+0.3$	V
Output Current		I_{OUT}	600	mA
Power Dissipation	SOT-23	P_D	400	mW
	SOT-353(SC70)		250	mW
	DFN1×1-4		400	mW
	SOT-89		600	mW
	TO-92		500	mW
Operating free air temperature range		T_A	-40~85	$^{\circ}\text{C}$
Operating Junction Temperature Range		T_j	-40~125	$^{\circ}\text{C}$
Storage Temperature		T_{stg}	-40~125	$^{\circ}\text{C}$
Lead Temperature(Soldering, 10 sec)		T_{solder}	260	$^{\circ}\text{C}$
ESD rating ⁽³⁾		Human Body Model - (HBM)	4	kV
		Machine Model- (MM)	200	V

(1) Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) All voltages are with respect to network ground terminal.

(3) ESD testing is performed according to the respective JESD22 JEDEC standard. The human body model is a 100 pF capacitor discharged through a 1.5k Ω resistor into each pin. The machine model is a 200 pF capacitor discharged directly into each pin.

■ RECOMMENDED OPERATING CONDITIONS

PARAMETER	MIN.	NOM.	MAX.	UNITS
Supply voltage at V_{IN}	2		6	V
Operating junction temperature range, T_j	0		125	$^{\circ}\text{C}$
Operating free air temperature range, T_A	0		85	$^{\circ}\text{C}$

■ ELECTRICAL CHARACTERISTICS

CE6200 Series

(V_{IN}=V_{OUT}+1V, C_{IN}=C_{OUT}=1μF, T_A=25°C, unless otherwise specified)

PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP. ⁽⁴⁾	MAX.	UNITS
Output Voltage	V _{OUT(E)} ⁽⁵⁾	I _{OUT} =1mA	V _{OUT} ⁽⁶⁾ *0.98	V _{OUT} ⁽⁶⁾	V _{OUT} ⁽⁶⁾ *1.02	V
Supply Current	I _{SS}	I _{OUT} =0		50	100	μA
Standby Current	I _{STBY}	CE = V _{SS}		0.01	1	μA
Output Current	I _{OUT}	—	500			mA
Dropout Voltage	V _{DO} ⁽⁷⁾	I _{OUT} =150mA V _{OUT} ≥2.8V		150		mV
Load Regulation	ΔV _{OUT}	V _{IN} = V _{OUT} +1V, 1mA≤I _{OUT} ≤100mA		10		mV
Line Regulation	$\frac{\Delta V_{OUT}}{V_{OUT} \times \Delta V_{IN}}$	I _{OUT} =10mA V _{OUT} +1V≤V _{IN} ≤6V		0.01	0.2	%/V
Output Voltage Temperature Characteristics	$\frac{\Delta V_{OUT}}{\Delta T \times V_{OUT}}$	I _{OUT} =10mA -40≤T≤+85		100		ppm
Short Current	I _{Short}	V _{OUT} =V _{SS}		100		mA
Input Voltage	V _{IN}	—	2.0		6.0	V
Power Supply Rejection Rate	217Hz	PSRR	I _{OUT} =50mA	80		dB
	1kHz			75		
	10kHz			70		
CE "High" Voltage	V _{CE} "H"		1.5		V _{IN}	V
CE "Low" Voltage	V _{CE} "L"				0.3	V
C _{OUT} Auto-Discharge Resistance	R _{DISCHRG}	V _{IN} =5V, V _{OUT} =3.0V, V _{CE} =V _{SS}		80		Ω

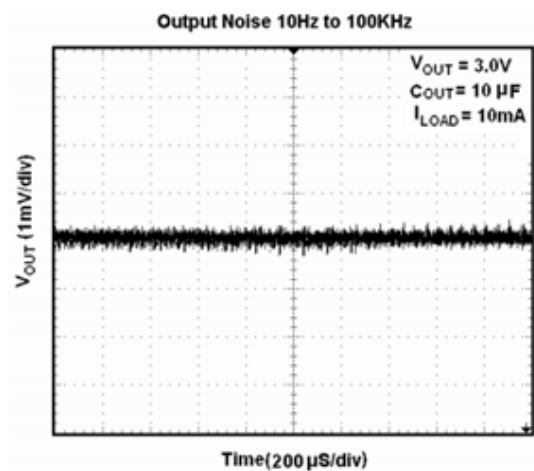
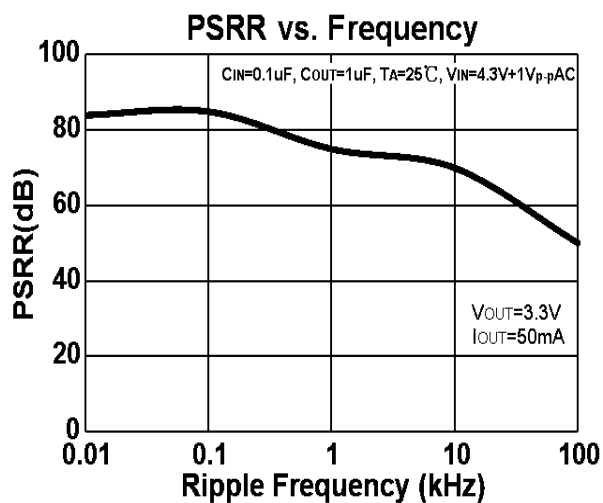
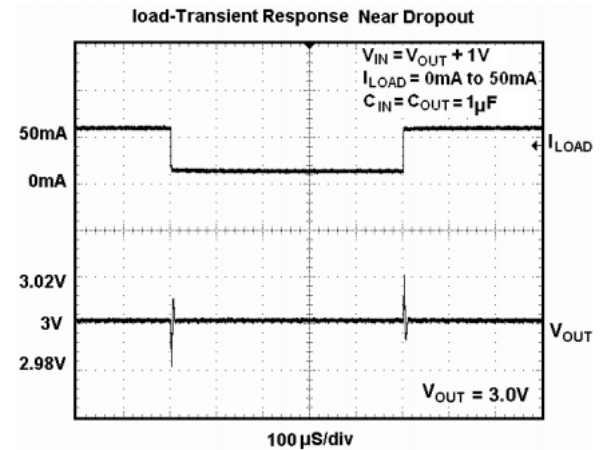
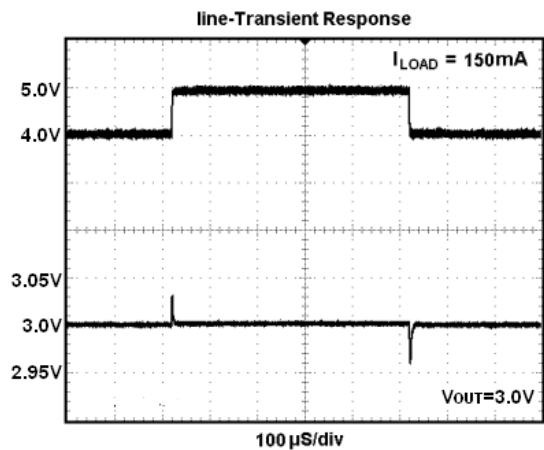
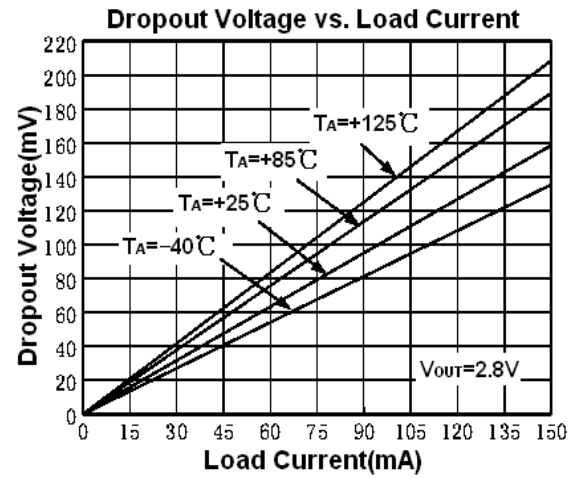
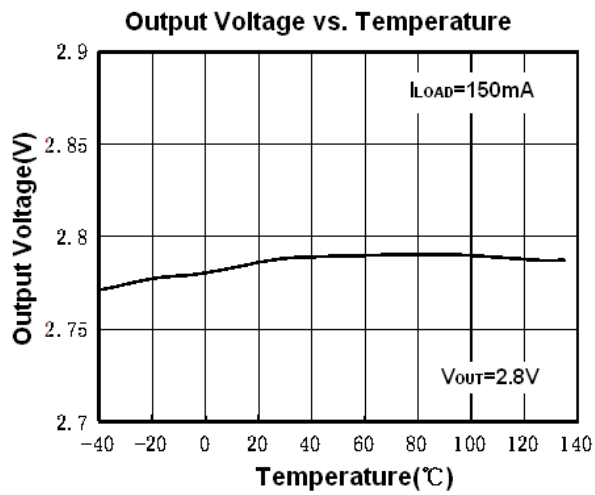
(4) Typical numbers are at 25°C and represent the most likely norm.

(5)V_{OUT(E)} : Effective Output Voltage (I.e. The output voltage when V_{IN}= (V_{OUT} +1.0V). And maintain a certain I_{OUT} Value).(6)V_{OUT}: Specified Output Voltage.(7)V_{DO}: The Difference Of Output Voltage And Input Voltage When Input Voltage Is Decreased Gradually Till Output Voltage Equals To 98% Of V_{OUT} (E).

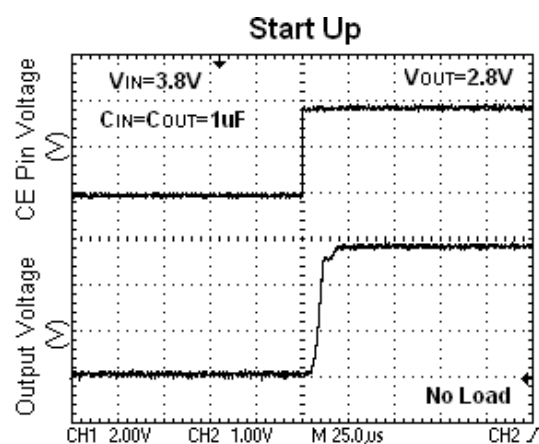
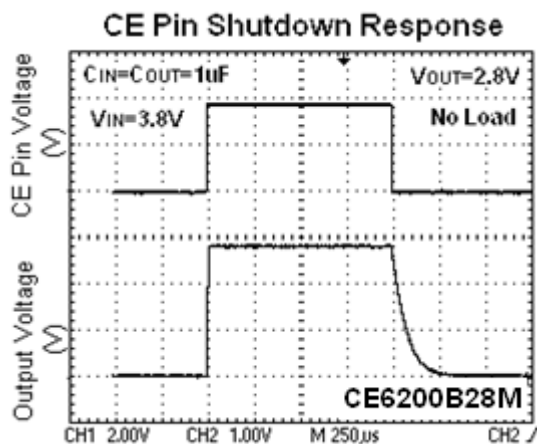
■ DROPOUT VOLTAGE CHART

Setting Output Voltage	Dropout Voltage(mV)@ I _{OUT} =150mA	
V _{OUT} (V)	Typ.	Max.
1.2	380	600
1.5	270	600
1.8	230	600
2.5	180	400
2.8	160	220
3.0	155	220
3.3	150	220

TYPICAL PERFORMANCE CHARACTERISTICS



■ TYPICAL PERFORMANCE CHARACTERISTICS



C_{OUT} Auto-Discharge Function

CE6200B series can discharge the electric charge in the output capacitor (C_{OUT}), when a low signal to the CE pin, which enables a whole IC circuit turn off, is inputted via the N-channel transistor located between the V_{OUT} pin and the V_{SS} pin (cf. BLOCK DIAGRAM). The C_{OUT} auto-discharge resistance value is set at 80Ω (V_{OUT}=3.0V @ V_{IN}=5.0V at typical). The discharge time of the output capacitor (C_{OUT}) is set by the C_{OUT} auto-discharge resistance (R) and the output capacitor (C_{OUT}). By setting time constant of a C_{OUT} auto-discharge resistance value [R_{DISCHRG}] and an output capacitor value (C_{OUT}) as τ ($\tau = C \times R_{DISCHRG}$), the output voltage after discharge via the N-channel transistor is calculated by the following formulas.

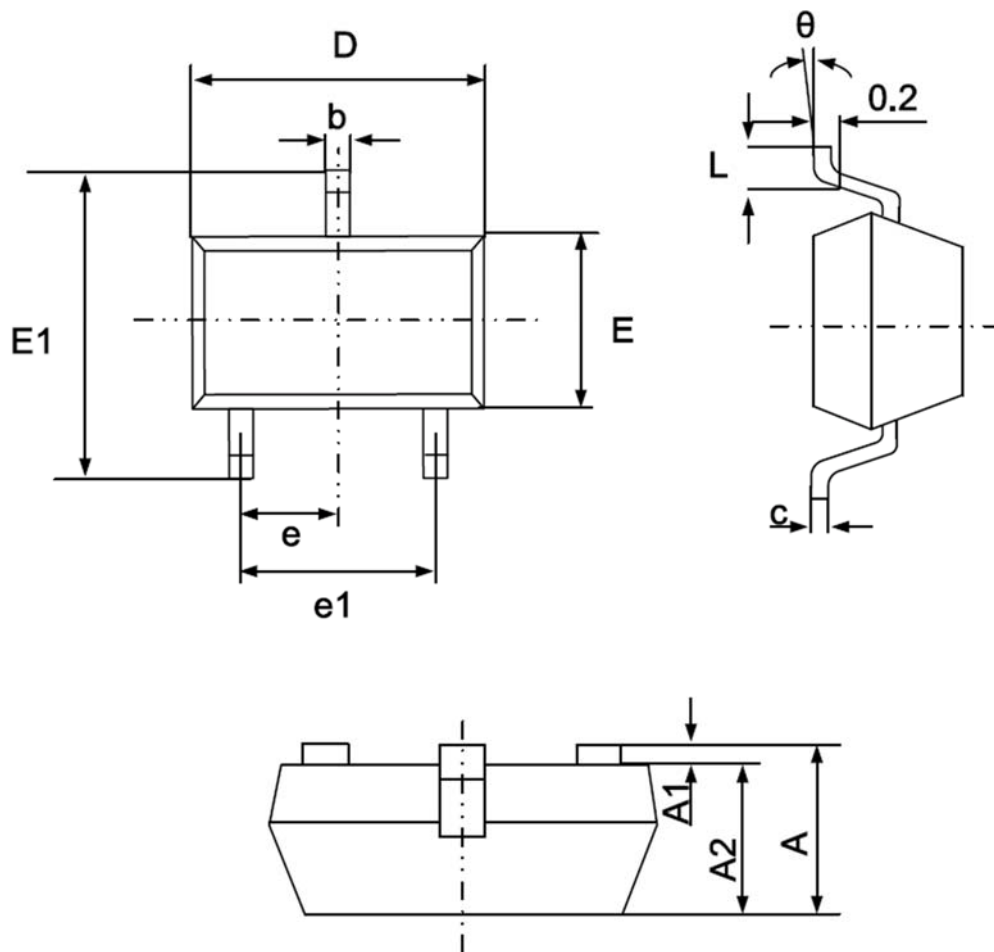
$$V = V_{OUT(E)} \times e^{-t/\tau}, \text{ or } t = \tau \ln (V / V_{OUT(E)})$$

(V: Output voltage after discharge, V_{OUT(E)}: Output voltage, t: Discharge time,

τ : C_{OUT} auto-discharge resistance R_{DISCHRG} × Output capacitor (C_{OUT}) value C)

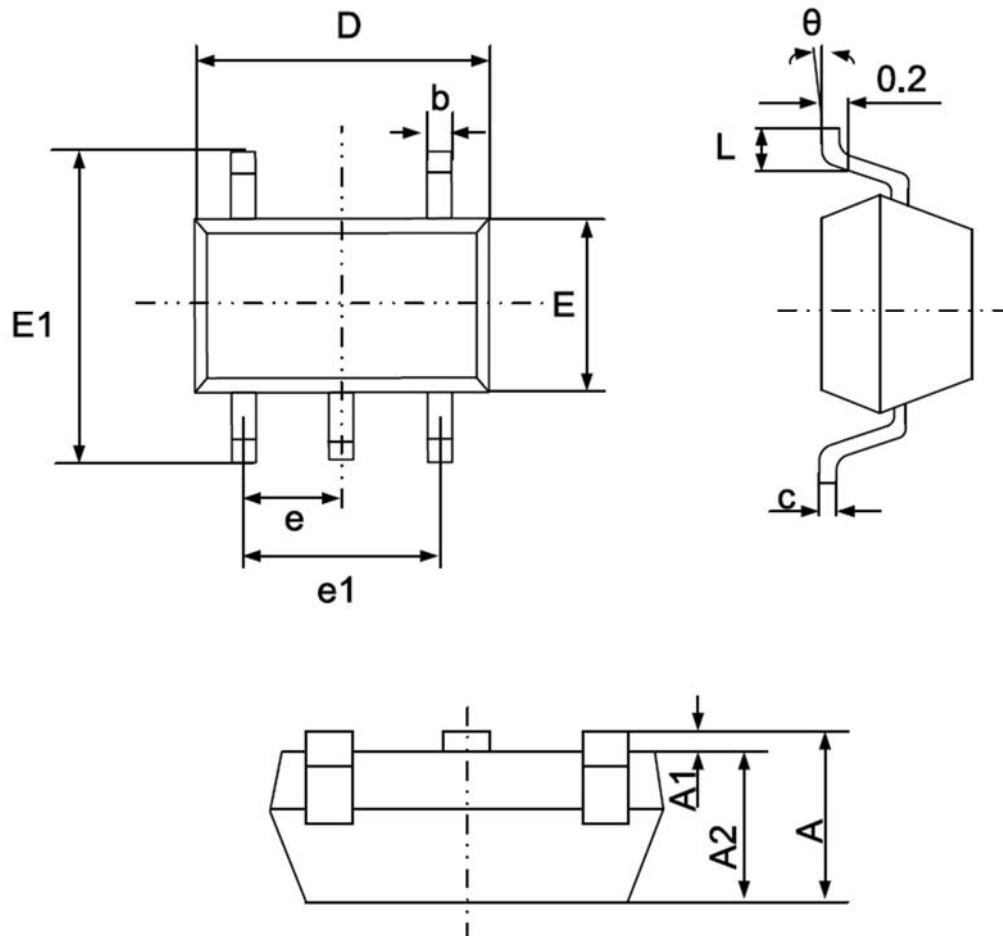
■ PACKAGING INFORMATION

• SOT-23-3 PACKAGE OUTLINE DIMENSIONS



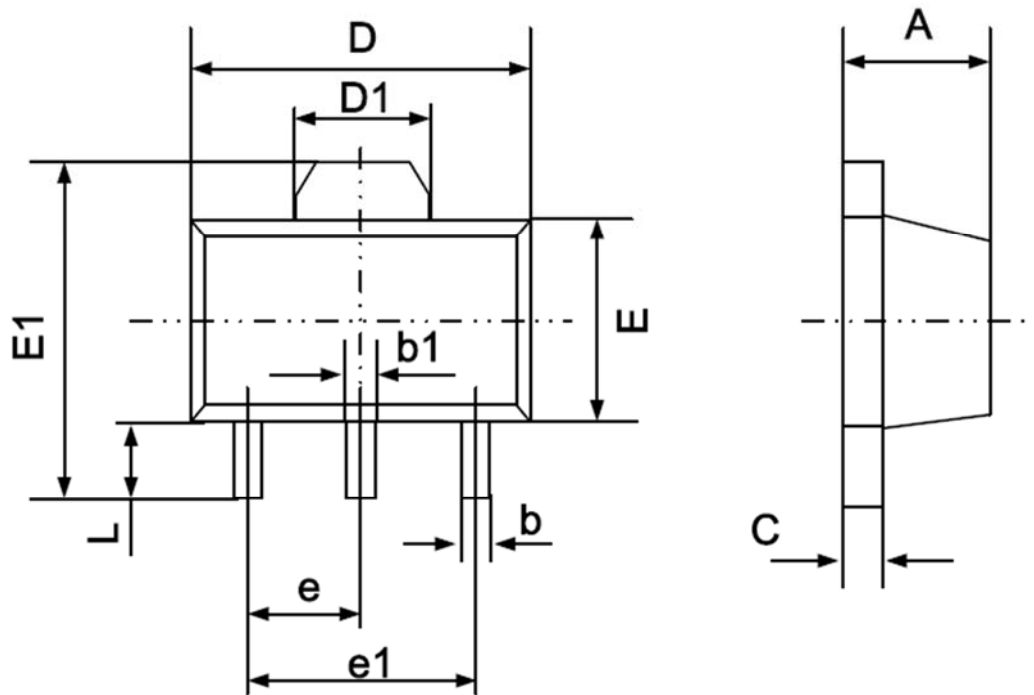
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

• SOT-23-5 PACKAGE OUTLINE DIMENSIONS



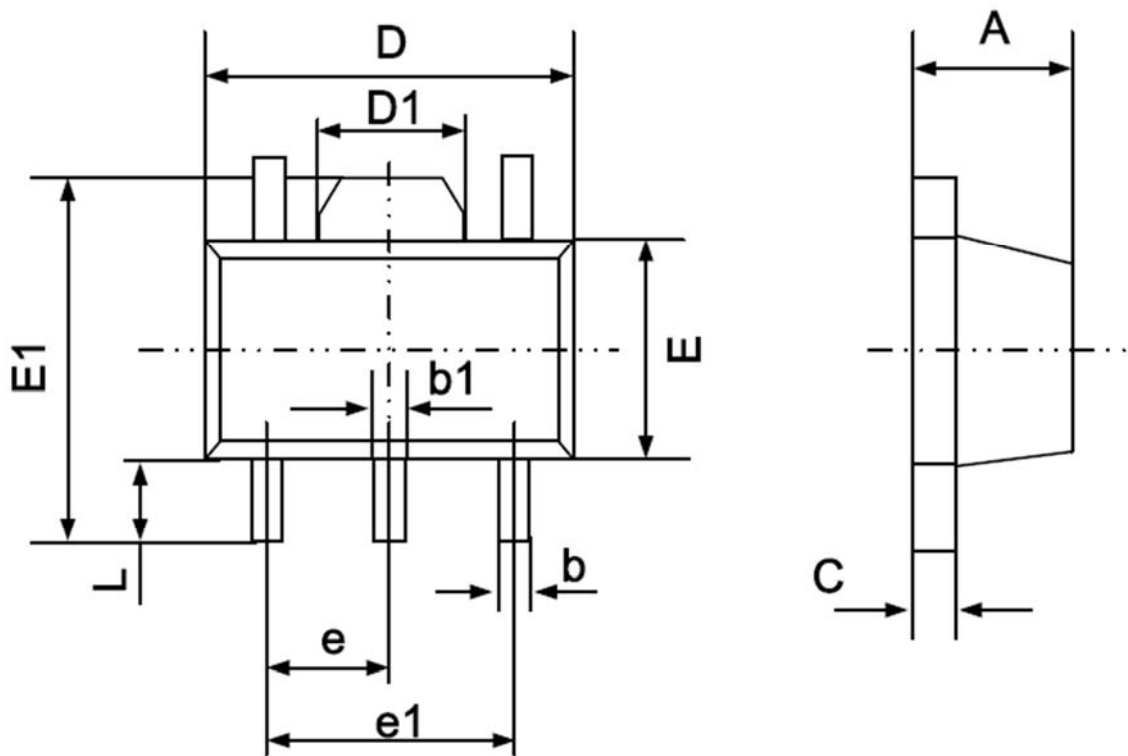
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

●SOT-89-3 PACKAGE OUTLINE DIMENSIONS



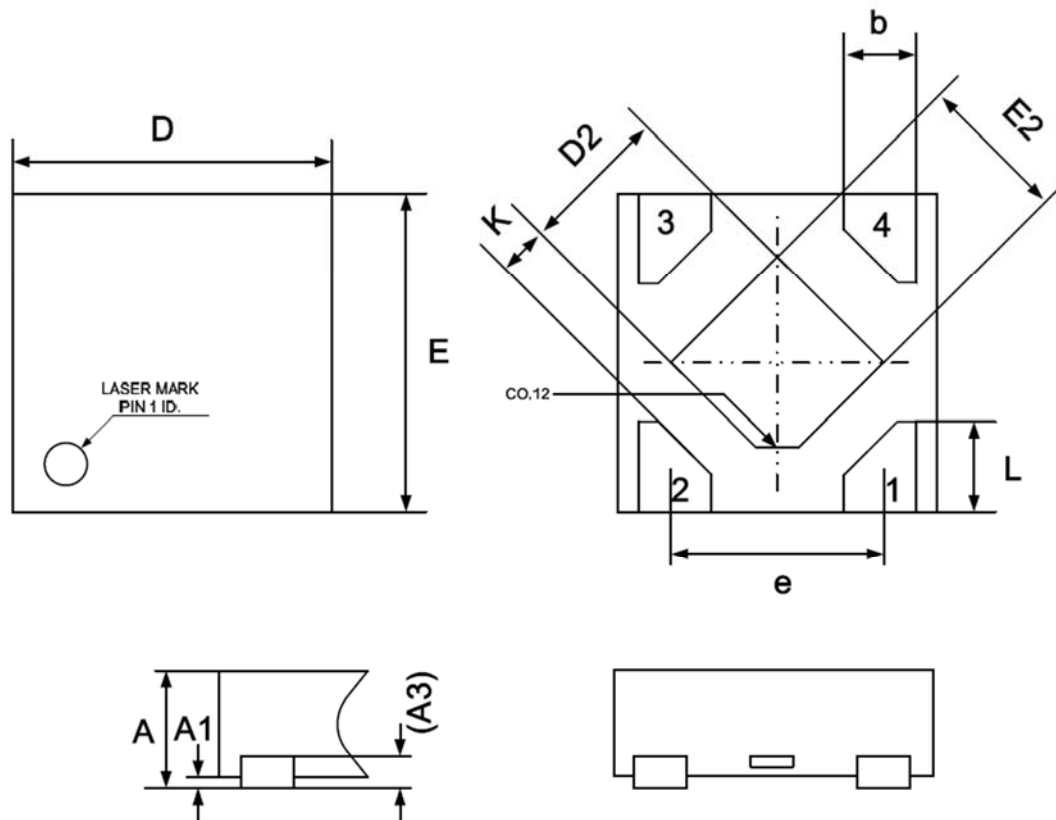
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF		0.061 REF	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060 TYP	
e1	3.000 TYP		0.118 TYP	
L	0.900	1.200	0.035	0.047

- SOT-89-5 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.360	0.560	0.014	0.022
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.400	1.800	0.055	0.071
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060 TYP	
e1	2.900	3.100	0.114	0.122
L	0.900	1.100	0.035	0.043

● DFN1×1-4 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		
	Min.	Nom.	Max.
A	0.340	0.370	0.400
A1	0.000	0.020	0.050
A3	0.100REF		
b	0.170	0.220	0.270
D	0.950	1.000	1.050
E	0.950	1.000	1.050
D2	0.430	0.480	0.530
E2	0.430	0.480	0.530
L	0.200	0.250	0.300
e	-	0.650	-
K	0.150	-	-

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